

Common Ports and Protocols — Guided Notes

Unit 1.5.1 | CompTIA Network+ N10-009 Obj. 1.5 | N10-008 1.5

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Ports are _____ -bit numbers ranging from _____ to _____. The three ranges are:

Well-known: _____

Registered: _____

Dynamic/Ephemeral: _____

Question 2

Fill in the port numbers:

FTP control / data → _____ / _____

SSH → _____

Telnet → _____

SMTP → _____

DNS → _____

Question 3

Fill in the port numbers:

DHCP server / client → _____ / _____

HTTP → _____

HTTPS → _____

SMB → _____

RDP → _____

Question 4

FTP transmits credentials in _____. Its secure replacements are _____ (runs over SSH, port 22) and _____ (FTP over TLS). Telnet is similarly insecure because _____.

Question 5

DNS normally uses _____ (TCP/UDP) on port 53 for standard queries because _____. It switches to _____ for zone transfers and large responses because _____.

Question 6

When your browser connects to a web server on port 443, your OS assigns a temporary _____ port (range: _____) as the source port. This port exists only for the duration of that _____.

Question 7

SMB uses port _____. The _____ ransomware in 2017 exploited a vulnerability in SMB version _____, which is why modern best practice is to _____.

Question 8 — Real World Application

REAL WORLD: A user can browse the internet normally but cannot connect to the company's internal file server or map any network drives. You run netstat on the file server and see it listening on port 445. What is the most likely cause and how do you confirm it?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-5-1-common-ports-and-protocols/1-5-1-common-ports-and-protocols-answer-key.pdf
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